



An initial user model design for adaptive interface development in learning management system based on cognitive load

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Abstract

The cognitive aspect is crucial in developing interactive and adaptive systems, including learning management systems (LMS). By understanding human cognitive processes, developers can create adaptive systems that are sustainably used by targeted users. Cognitive traits need to be explored to become part of the user model, especially especially to develop an LMS with an adaptive interface based on cognitive load. However, there is limited research exploring the correlation between cognitive load and adaptive interfaces, as well as the visualization of information, within online learning settings, including those specifically within LMS platforms. Additionally, a lack of understanding of cognitive processes renders the use of LMS static and unable to adapt to learners' abilities. Therefore, information regarding the specific characteristics or model of cognitive load that accurately represent LMS users as triggers for adaptive interfaces needs to be further examined. A total of four experts in the cognitive field were interviewed. The interview process was conducted in a semi-structured manner to obtain information about problems in the use of e-learning, the nature of cognitive, and the process of capturing cognitive load during learning using LMS. By using a soft system methodology, the results from the interviews are mapped into a cognitive load-based LMS user model design. The initial model design includes information on working memory capacity, task performance (time, true/false rate), learning behavior (learner log), physiological (in a lab scale), and self-reporting (Likert scale). This model design is a promising first step towards a more technical process in developing adaptive and interactive systems.

Keywords Adaptive user interface · Cognitive load · Learning management system · Soft system methodology · User model

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1 Introduction

Acceleration in the field of education is necessary in response to the increasing need for fast learning and availability at any time, aiming to foster creativity and innovation in the industrial era 4.0 (Gowripeddi et al. 2021). Specifically, in Indonesia, has been emphasized by the need for sustainable learning activities since the COVID-19 outbreak, which necessitated fully online or blended learning modes (Chaeruman and Maudiarti 2018). To support these learning activities, the implementation of LMS technology has become essential.

LMS is an application that contains features necessary for the learning process. Teachers can conduct classes and exchange information interactively with students, and access to learning materials is available at preset times (Maliki et al. 2021). According to BuiltWith, there are at least 6,315 websites in Indonesia that include LMS by the end of May 2022, with 59% of them being Moodle-based (BuiltWith Company 2022). Most LMS such as Moodle, Claroline, Dokeos, Docebo, ATutor, Chamilo, Edmodo, Schoology, and OLAT are web-based with full features, including user management, course and learning material management, activity and grade management, and elearning visualization management. These features are supported by abundant menu displays that offer settings according to user needs (Ramos Landeros and Margain Fuentes 2016).

Despite providing many benefits, the features and appearance of LMS settings also introduce complexity for users. From the teachers' perspective, technological competence remains a problem, causing technology anxiety (Bervell and Umar 2020). Moreover, teachers find it challenging to facilitate online learning effectively in LMS, often feeling that limited direct interaction time hampers their ability to explain material in detail, gauge students' emotions, and manage students' affective states during learning. Consequently, teachers store abundant materials in LMS, hoping they can be fully accessed by students anytime and anywhere (Rosalina and Elyani 2020).

From the students' perspective, LMS often fails to encourage learning and lacks a good monitoring and management system. This necessitates real teachers to supervise, facilitate, and complete the learning process (Zhao 2018). Teachers frequently provide large quantities of material, causing cognitive overload. Although research by Chang and Ley (2006) indicates that printing some learning materials can mitigate cognitive overload, it does not reduce extraneous cognitive loads such as complex displays, excessive information, and efforts to link information in online learning.

According to Chen et al. (2016), cognitive load is a multi-dimensional dynamic construct representing the working memory load during cognitive tasks. Sweller

(1994) categorizes cognitive load into intrinsic cognitive load (ICL), extraneous cognitive load (ECL), and germane cognitive load (GCL). In online learning, students' cognitive load tends to increase. Consequently, many researchers research to reduce cognitive load, especially extraneous cognitive load related to the presentation of online material forms (Costley 2020; Rahmat et al. 2022; Warrick 2022). However, very few studies have examined the relationship between cognitive load and adaptive interfaces and visualization of information in online learning environments, specifically within LMS.

Therefore, to develop an adaptive interface in LMS based on the users' cognitive load status, understanding the LMS user's cognitive load model needs to be strengthened. This qualitative research focuses on extracting information from educational psychologists regarding cognitive load and its relation to LMS usage, aiming to discover the initial cognitive load-based LMS user model design.

The research question addressed is: *"What does the initial cognitive loadbased LMS user model design look like?"* This question is answered through:

1. Extracting information about individual differences in students to minimize bias in cognitive load-based user modeling.
2. Obtaining information from experts on relevant tasks and methods to detect user cognitive load at early, during, and late learning stages.
3. Drawing conclusions about data types from the detection process that can represent a cognitive load-based LMS user model.

The proposed cognitive load-based LMS user model aims to measure fluctuations in user cognitive load using subjective-only, objective-only, or a combination of both approaches. This model serves as a guideline for collecting data from respondents to detect cognitive load fluctuations and inform adaptive user interfaces and learning strategies.

The article is organized as follows: Sect. 1 describes the study's background. Section 2 reviews related theories on user interfaces, adaptive user interfaces, and human cognition. Section 3 explains the study's methodology. Section 4 discusses the results and findings. Finally, Sect. 5 presents the conclusions and directions for future research.

2 Relevant literature review

This section reviews theories that support the development of an adaptive user model based on deep cognitive load for adaptive user interfaces. The related theories encompass user interface concepts, adaptive user interfaces, and human

cognition, focusing on information processing theory, cognitive load, Cognitive Load Theory (CLT), and cognitive processes in Learning Management Systems (LMS). Additionally, recent related studies are reviewed to understand the basics of human cognition, align with expert insights, and explore the potential for developing a cognitive load-based user model for adaptive user interfaces.

2.1 User interface and adaptive user interfaces

According to Galitz (2007), the user interface (UI) as the component of computers and software that enables human interaction with machines, involving processes like seeing, hearing, touching, and speaking via input and output devices. Over the past three decades, interface development has advanced significantly, with at least 20 common types in today's digital society (Sharp et al. 2019).

A well-designed user interface is often assessed by its usability. According to ISO 9241-11, usability measures the extent to which a product or service can be used by specific target users to achieve their goals effectively, efficiently, and appropriately in the given context (Bevan et al. 2016). Consequently, the user-centered design (UCD) approach is frequently employed in UI development. ISO 9241 – 210:2010(E) outlines four UCD stages: understanding and determining the context of use, identifying user requirements, producing design solutions, and evaluating the design against requirements.

Crestani et al. (2017) categorize user interface development activities into four abstraction levels: conceptual, semantic, syntactic, and lexical, from highest to lowest. Various user interfaces, such as command language interfaces, menus, forms, natural language, direct manipulation, augmented and virtual reality, and combinations, are implemented. Design principles based on theory, experience, and common sense guide interface development to ensure ease of use.

Design principles focus on what to include and exclude in the interface. Sharp et al. (2019) propose five principles: visibility, clear feedback, defined boundaries, logical nature, and consistency. Gordon (2020) suggests that design elements like lines, shapes, colors, grids, and spaces can be combined to enhance visuals and user experience.

User interfaces are evolving towards adaptivity, producing intelligent interfaces that depend on human-system interactions. An adaptive user interface (AUI) identifies interaction patterns and adjusts the interface based on specific rules or models (Bagustari and Santoso 2019). Riascos et al. (2017) emphasize the importance of user modeling, adaptive engines, and interaction repositories in AUI development frameworks for various applications.

The user modeling component for AUI needs to be considered in detail because it will be a trigger for the intelligence processes. Other researchers have proposed various approaches such as prior knowledge (Ciolacu and Beer 2016); ontology-based to model user behavior (Iqbal et al. 2021); cognitive load measurement specifically for users with cognitive disabilities (Bend et al. 2021; Gajos and Chauncey 2017; Heumader et al. 2020; Riascos et al. 2017). In the adaptive engine component, AUI involves simple machine learning methods to deep learning. The Gravitational Search based Back Propagation Neural Network (GSBPNN) method, the Back Propagation Neural Network (BPNN) method (Kolekar et al. 2017), Fuzzy C Means (Kolekar et al. 2018), rule-based (Kolekar et al. 2019), and Reinforcement Learning (Todi et al. 2021). Meanwhile, in the delivery of interfaces resulting from adaptation, many researchers focus on filtering information and sequencing learning materials and activities (Ciolacu and Beer 2016; Kolekar et al. 2018; Maaliw 2020).

2.2 Human cognition

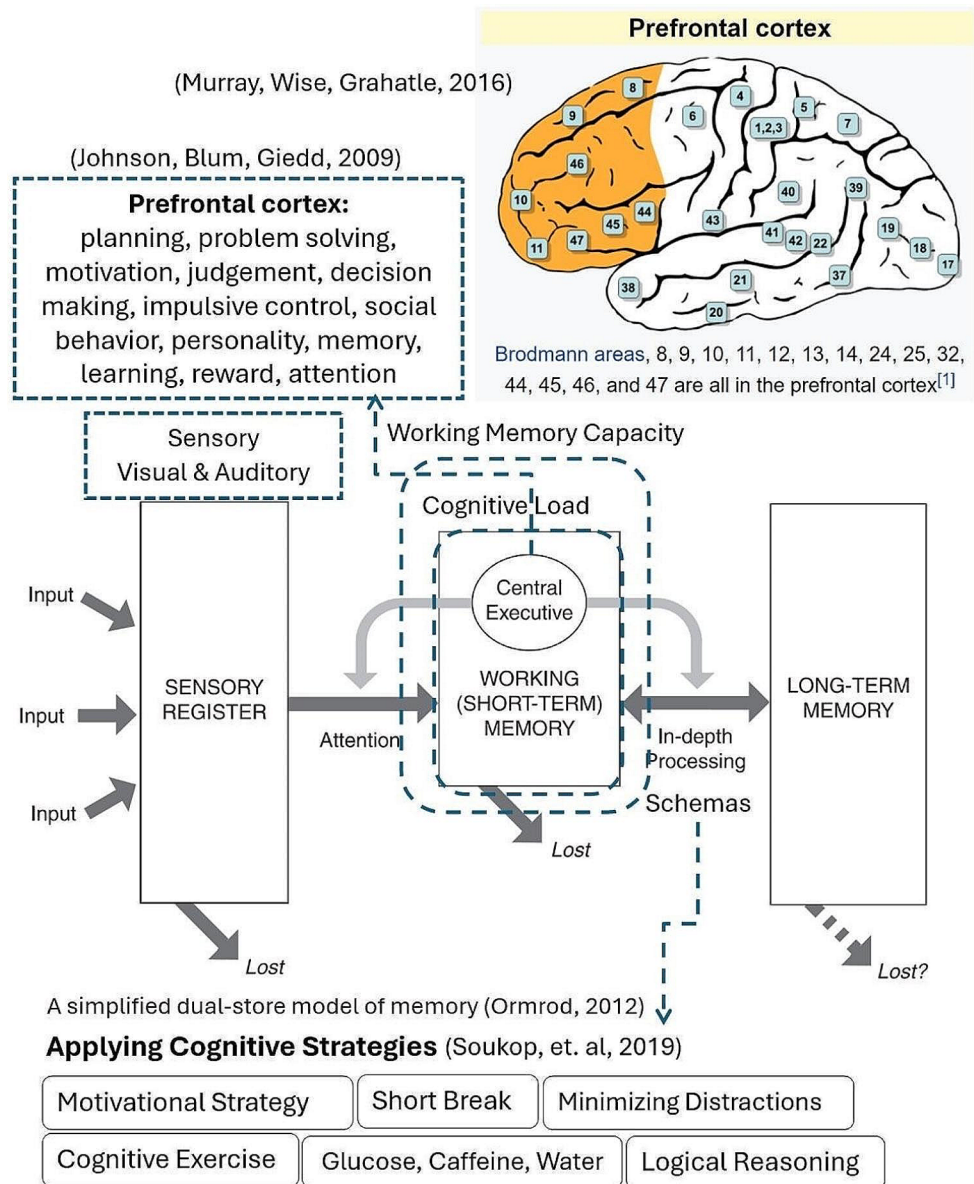
Cognition involves mental processes for perceiving, remembering, and thinking about events via auditory and visual sensors, essentially how humans learn or know (Omrod and Ormrod 2012). Cognitive performance depends on cognitive factors related to attention, memory, and reasoning (Lovejoy 2013). Understanding information processing theory is essential to detect cognitive load in users and tailor adaptive processes. In detail, the reviews related to cognition is presented as follows.

2.2.1 Information processing theory

Information processing theory is concerned with how the process of stimuli in the form of information received from the environment and then storing it in memory. Then the information is retrieved from memory to be conveyed or used to complete other cognitive tasks (Omrod and Ormrod 2012). An understanding of information processing theory is important to later support the cognitive load detection process in users. In addition, it can also determine the focus of giving treatment or an adaptive process.

Figure 1 describes information processing theory and the role of memory as a dual-store model (Omrod and Ormrod 2012). Stimulus information either in visual or auditory format is entered as input received by human sensory. Human visual and auditory senses are strongly influenced by attention. Attention is the initial gate for information to enter and be processed in short-term memory. Attention involves selecting some information to be processed further or to be blocked from being processed further (Kellogg 2016).

Fig. 1 The information processing theory in the dual-store memory model, its correlation with human cognition in the prefrontal cortex, and cognitive strategies that can be implemented. *Source:* Adopted from mentioned references and redraw by authors



Information that is successfully entered from the attention filter is then processed in short-term memory. Short-term memory (STM), also referred to as primary or active memory, also called short-term storage, denotes the different memory systems involved in storing chunks of information for a relatively short period, usually up to 30 s (Cascella and Al Khalili 2019). Short-term memory serves as a temporary recall for a limited amount of information from the sensory register for processing after attention or recognition. In verbal form, George Miller conveys the term magical number, which is 7 ± 2 the amount of information that can be recalled at one time (Sweller 1994). Short-term memory is also known as working memory and every human being has their own limited working memory capacity. Working memory capacity (WMC) is seen as individual differences

which can be categorized into three types, namely: the executive-attention view of WMC, the primary-memory/secondary-memory view, and the binding hypothesis. These three perspectives on WMC can be measured by different psychological tasks (Wilhelm et al. 2013).

In short-term memory, there is an executive function. An executive function is a collection of top-down control processes that are automated or rely on instinct or intuition to be wrong, inadequate, or impossible (Diamond 2013). The executive function processes include planning, problem-solving, motivation, judgment, decision making, impulsive control, social behavior, personality, memory, learning, reward, and attention (Johnson et al. 2009). Physically in the brain, the executive function process is in the prefrontal cortex. The prefrontal cortex is used to determine changes

in cognitive abilities non-intrusively through brain wave detection (Ahmed et al. 2020; Huang et al. 2020a; Kuanar et al. 2018; Maimon et al. 2021; Plechawska-Wojcik et al. 2018), especially observations in the Brodmann areas, 8, 9, 10, 11, 12, 13, 14, 24, 25, 32, 44, 45, 46, and 47 brain area (Murray et al. 2017).

Furthermore, after the information is processed in short-term memory, our brain system performs in-depth processing intending to store the information in long-term memory. Long-term memory is capable of storing an unlimited amount of information (Cascella and Al Khalili 2019). Generally, to streamline the process of storing information in long-term memory and to make it easier to retrieve that information, various cognitive strategies are implemented. Soukup et al. (2019) said that several cognitive strategies can be used, such as short breaks of 5–10 min for light physical exercise, regulating glucose and caffeine consumption, doing cognitive exercises, practicing logical reasoning, minimizing distractions to motivational strategies.

2.2.2 Cognitive load and cognitive load theory

Working memory has a separate processor to handle visual and verbal information, as noted by Baddeley (1992). It can process a limited amount of information at one time for a short duration (Cowan 2001). When human perform complex tasks or engage in learning, it requires a large amount of working memory, which is referred to as cognitive load.

Cognitive load refers to the demand on working memory caused by a complex task that requires new information or requests to be processed (Sweller et al. 1998). The cognitive load experienced by each person when dealing with the same task can vary. This variation can be due to several factors, such as prior knowledge, familiarity with the interface, age, or physical or mental disabilities. A task that is difficult for one person may not be difficult for someone with more experience (Chen et al. 2016).

The limitation of working memory in processing information forms the basis for considering appropriate instructional design through cognitive load theory. Cognitive load theory suggests that the primary goal of teaching should be to increase knowledge that is beneficial to long-term memory. Therefore, information needs to be presented in a way that reduces unnecessary processing so that cognitive load can be maximized for processing important new information.

Cognitive load theory is a constructivist theory that helps instructors develop learning media aligned with the limitations of human cognitive architecture. It is categorized into three types: intrinsic cognitive load (ICL), extraneous cognitive load (ECL), and germane cognitive load (GCL). Intrinsic cognitive load relates to the inherent complexity

of the learning material itself and cannot be avoided. Extraneous cognitive load refers to the complexity of the way information is presented. The assumption is that extraneous cognitive load can be minimized to allow more processing capacity for intrinsic cognitive load. Germane cognitive load refers to the cognitive resources used for constructing new schemas in long-term memory (Paas and Van Merriënboer 1994).

Intrinsic and extraneous cognitive loads are additive. These two types of cognitive load determine the total cognitive load influenced by the material that needs to be studied. In creating instructional designs, extraneous cognitive load is reduced as much as possible to strengthen the relationship between intrinsic cognitive load for learning information and to facilitate schema development through germane cognitive load. The degree of intrinsic and extraneous load is determined by the interactivity of the elements. Interacting elements are elements that must be processed simultaneously in working memory because they are logically related.

The interactivity of the elements can be estimated for any information the student needs to learn. Regarding intrinsic load, some materials can be studied one element at a time and are thus low in element interactivity and intrinsic cognitive load. High-element interactivity materials require more working memory resources than materials with low-element interactivity. All relevant elements must be processed simultaneously to be studied meaningfully (Sweller et al. 2011).

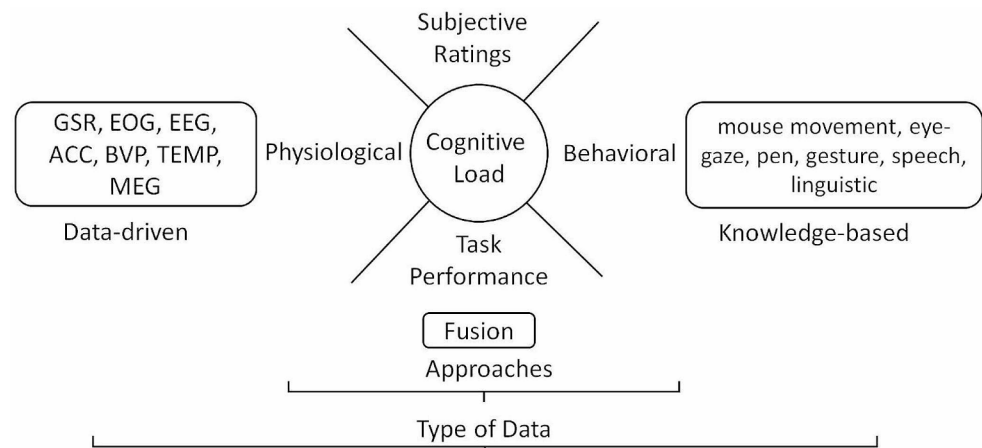
2.2.3 Measuring cognitive load

In the context of developing interactive and adaptive systems, especially intelligent ones, the user's cognitive load needs to be considered. The ability to pinpoint exactly when users are cognitively loaded beyond manageable levels allows the system to intelligently adjust its interaction strategy. Within the scope of HCI and other domains, many researchers have attempted to measure users' cognitive load. In general, there are four main categories of methods for measuring cognitive load: subjective measures, performance measures, behavioral measures, and physiological measures (Chen et al. 2016). An overview of cognitive load measurement methods is depicted in Fig. 2.

According to O'Donnell and Eggemeier (1986), subjective measures or self-reports are traditional methods with the most consistent results. Users are asked to perform certain activities first, then at the end of the activity, they reflect by filling out standardized questionnaires. There are two types of scales in this method:

1. Unidimensional scales: These measure overall cognitive load, such as the Subjective Cognitive Load Measurement Scale (Paas and Van Merriënboer 1994) and

Fig. 2 An overview of various methods of cognitive load measurement. (GSR: Galvanic Skin Response, EOG: Electrooculography, EEG: Electroencephalogram, ACC: Anterior Cingulate Cortex signal, BVP: Blood Volume Pulse, TEMP: Temperature, MEG: Magnetoencephalography. *Source:* Adopted from Chen et al. (2016). and redraw by authors



the subjective one-item measure modified from NASA-TLX proposed by von Janczewski et al. (2022).

2. Multidimensional scales: These focus on measuring different components of cognitive load, namely intrinsic, extraneous, and germane cognitive load. Examples include NASA-TLX developed by Hart and Staveland (1988), the Three-item scale proposed by Cierniak et al. (2009), and the ten-item subjective cognitive load scale developed by Leppink et al. (2013).

Although the results of these subjective measures are consistent, there are limitations. This method is implemented after the activity is finished, so the user's cognitive load during the learning process cannot be detected (Paas and Van Merriënboer 1994). Filling out questionnaires often interrupts the flow of tasks and adds to the workload for users who may already be overloaded (Chen et al. 2012).

The next method is performance measures, which can reflect variations in cognitive load during the task. If a user's performance decreases or they stop performing the task, it can be interpreted that the user has a high cognitive load or is overloaded (Paas and Van Merriënboer 1994). Performance measures are related to task completion time and true/false rates (Ejaz et al. 2019; Modanwal et al. 2022). The correlation between performance and cognitive load can be categorized as Low, Medium, and High (O'Donnell and Eggemeier 1986). However, this method cannot measure the user's spare capacity when they still have enough working memory to perform the task (Parasuraman et al. 2008). Additionally, this method relies heavily on task completion, which can become invalid if the user is stuck on initial steps for a long time.

Another method closer to the user's behavior is behavioral measures. This method observes patterns of interactive behavior when the user performs a task. Examples include observing eye-gaze (Evans and Fendley 2017; Pillai et al. 2022), mouse pointing and clicking (Chen et al. 2012), gait patterns (Martin and Bajcsy 2011), and digital pen input

(Ruiz et al. 2007). From these responses, two pieces of information are obtained: the meaning of the response and the way the response was made. However, the need for consistency, compact representation, and automatic acquisition is extremely high so that the data obtained can be processed and implemented in an adaptive system (Chen et al. 2016).

The most sophisticated measurement methods are physiological measures. This method assumes that changes in human cognitive functioning can be reflected in physiology (Ayres et al. 2021). The advantage is the certainty of obtaining bodily data continuously and detecting cognitive load with high sensitivity. Several physiological aspects measured include brain activity through electroencephalography (EEG) signals (Ahmed et al. 2020; Huang et al. 2020a), electrocardiography (ECG), and heart rate (Hettiarachchi et al. 2019). Other data sources include eye activity such as electrooculography (EOG) (Belkhiria and Peysakhovich 2021) and pupillary dilation (Toker & Conati, 2017), and Galvanic Skin Response (GSR) (Hossain et al. 2019). However, like other methods, physiological measures have limitations such as the need for non-intrusive devices, costs of tools, and the need for experts to interpret data. Additionally, the sensitivity and variability of physiological signals can be influenced by external factors like temperature and movement, which may not be suitable for daily use (Delis et al. 2001).

The high demand for developing interactive and adaptive applications based on artificial intelligence, involving human cognitive load and task complexity, makes it insufficient to use only one modality. Recently, a combination of cognitive load measurement methods called multimodal cognitive load measurement (MCLM) has emerged (Chen et al. 2016). Figure 2 also shows that cognitive load measurements can be categorized into data-driven for physiological methods and knowledge-based for behavioral methods. The different modalities of the four categories can be combined to perform multimodal cognitive load measurement. The involvement of applications, engineering processes, or tools

such as machine learning is used to process data from this strategy.

2.2.4 Cognitive process in learning management system

The development of adaptive interface systems based on cognitive load has been extensively researched. However, specific implementations in the education domain, particularly in learning management systems (LMS), are still rare. In Lin and Graf (2007), Divergent Associative Learning (DAL) was introduced as an aspect of cognitive traits to model learner profiles. In this research, DAL was used as a new component to complement working memory and inductive reasoning. Divergent Associative Learning (DAL) itself is a learning characteristic that develops relationships between new concepts and those that have already been studied. The DAL characteristics include associative hierarchy, classifiability, versatile navigation, relevance filtering, domain performance, and working memory. The effectiveness of DAL as part of the Cognitive Traits Model (CTM) was tested using LMS by users. Users were asked to access the LMS, starting from logging in to completing quizzes. The fluency variable obtained from the number of questions answered and the original score, termed *sumOrg*, was used for correlation analysis. The two variables were positively correlated, indicating that DAL can be used as one of the characteristics of the Cognitive Traits Model.

The concept of the Cognitive Traits Model (CTM) used by (Lin and Graf 2007) served as a reference for Lwande et al. (2021) to develop an LMS based on learning style and cognitive traits to personalize learning. In this research, the CTM concepts used were associative learning, working memory capacity, inductive reasoning ability, and information processing speed. These four concepts were mapped into activities in the LMS such as accessing learning objects, forward and backward navigation, accessing examples, revision, and exercises. To identify cognitive traits, a cognitive trait estimation process was carried out using specific rules and formulas. The results were mapped to a three-item scale: 0.1–0.3 (low), 0.4–0.6 (moderate), and 0.7–1.0 (high) to determine the appropriate preferences.

In another study conducted by Llantos and Estuar (2018), efforts to reduce cognitive load, referred to as mental effort, were focused on maintaining a fast response system for handling requests. Using rules adopted from Dabrowski and Munson (2011), several categories of ideal responses were established. The fastest response should be given by the system within 100–200ms when the user is typing and clicking. The system must understand user actions within 0.5–1 s. To maintain continuous interaction, the response should be within 25 s, and to keep the user engaged, the response should be within 7–10 s. To achieve this, the researchers

optimized infrastructure using cloudbased server services. However, there have been no further studies related to the effectiveness of reducing cognitive load through this approach.

Previous research has focused on higher cognitive load aspects such as working memory capacity, inductive learning, associative learning, and divergent associative learning. However, in the context of cognitive load theory (CLT), the application of CLT to LMS usage activities, both in terms of interface and access to material content, is still underexplored. The focus on cognitive load theory in LMS can be further studied.

3 Methodology

In this study, the basic understanding of theories related to human cognition forms the foundation for extracting further information from experts. The information extraction process utilizes the scientific framework of Soft System Methodology (SSM), proposed by Checkland in 1981. According to Avison and Fitzgerald (2003), there are several considerations for using SSM, including:

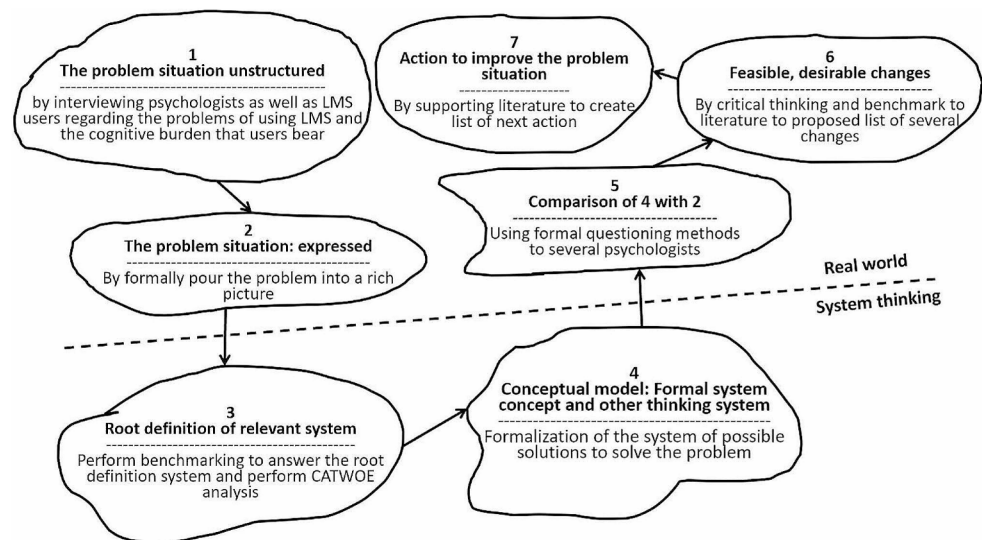
1. SSM can facilitate different interpretations of problems by relevant stakeholders.
2. The human factor is an important consideration.
3. SSM is creative and intuitive as a problem-solving tool, emphasizing the study of existing problems rather than direct practical solutions.

According to Fig. 3, there are seven stages of SSM implemented in this study, detailed as follows:

Stage 1: The problem situation - unstructured. In alignment with the research question on developing a cognitive load-based LMS user model design, experts in the cognitive field are needed. Therefore, this stage involves identifying experts and extracting information through semi-structured interviews. The discussions cover aspects such as the nature of cognition, issues regarding the detection of cognitive load in the learning process, and other challenges encountered when using LMS.

Stage 2: The problem situation - expressed. At this stage, interviews are conducted in more detail, supported by the search for documents or policies that might affect LMS implementation by educational institutions. The interview results are processed through coding and triangulation before being visualized in the form of a rich picture. This visualization helps understand the problem description of using LMS, focusing on the user's cognitive load, before proposing a solution.

Fig. 3 Research methodology that adopts from Soft System Methodology (SSM) Model 1 by Checkland. *Source:* Draw by author, adopted from Avison and Fitzgerald (2003)



Stage 3: Root definition of the relevant system. Benchmarks are carried out for the relevant system, and questions related to what, how, and why are addressed. CATWOE analysis from Bergvall-Kåreborn et al. (2004) is conducted to strengthen the root definitions of the relevant system that may serve as a solution.

Stage 4: Conceptual model. Based on the root definition of relevant systems, critical thinking, and coding interview results, a conceptual model of a potential system solution is developed. Triangulation of interview results is key to developing this conceptual model. Stage 5: Comparing the conceptual model with the problem situation. The discussion with experts is deepened by comparing different aspects of systems or potential conflicts when implemented in the real world. Solutions or supporting references are sought for the differences identified. The comparison technique used is formal questioning as referenced by Ebrahimi (2020).

Stage 6: Feasible, desirable changes. Both the agreed-upon implementation process for user model development and differing approaches with found reference solutions are inventoried. Alignment between expert statements and supporting references is also carried out.

Stage 7: Action to improve the problem situation (Implication). At this implication stage, the author lists potential actions needed to build a solution and explores areas that require further investigation.

4 Results and discussion

To answer the research questions, the three efforts mentioned in the introduction were distributed across the seven stages of SSM. The detailed research results are as follows.

Table 1 Information about psychologists involved in research

No	Initial	Gender	Qualification
1	MW	Female	Academic resilience, emotional competence, career development, and cognitive
2	WY	Male	Intelligence and cognition, wellbeing, vocational interest, hoax, psychometrics, and test construction
3	RR	Female	Intelligence and cognition, an experiment in cognition
4	HS	Male	Student wellbeing, achievement emotion, measurement tool development, critical thinking, and cognitive ability

As the experts were still recovering from COVID-19 and for ease of interaction, the interview process was conducted online using video conferencing tools. The duration of each interview ranged from 45 to 60 min. For additional confirmation purposes, the experts were willing to be contacted via email or instant messaging apps. The list of questions asked was derived from literature studies related to cognitive load and LMS research. The interview questions are as follows:

4.1 The result of stage 1: The problem situation unstructured

At this stage, four cognitive experts were interviewed. Detailed information about the four educational psychologists involved in this study can be found in Table 1. In addition to being experts in the cognitive field, all four are lecturers in the Faculty of Psychology at Padjadjaran University, Indonesia. This is an added advantage as these experts also have an understanding of university-level learning conditions.

1. Do you use LMS in your teaching? What obstacles do you and your students generally experience?
2. What does cognitive load mean?
3. What needs to be considered from the perspective of LMS users in measuring cognitive load?

4. How do measure a person's cognitive load at the beginning before studying? What tasks are best to use?
5. What are the techniques/methods for capturing cognitive load during the learning process?
6. Is there a specific time reference for measuring cognitive load in the learning phases?
7. In the context of using LMS, at which stages/cases is it possible to increase cognitive load?
8. From the user interface point of view, do the color, layout, and text properties affect cognitive load?

The results of this first stage included a list of problems regarding cognitive load detection and the use of LMS. This list was obtained from processing interview transcripts and triangulation. Formally, the results of stage 1 activities are modeled in a rich picture in stage 2.

4.2 The result of stage 2: The problem situation expressed

Based on the results of interview triangulation in stage 1, the author developed a rich picture that provides a broad overview of the existing problems. An overall depiction of user problems when using an LMS, as reported by various stakeholders, can be seen in the rich picture shown in Fig. 4. In Indonesia, policies from the Ministry of Education regarding the use of LMS as a learning support are in place. The use of LMS will be evaluated and will contribute to the clustering of higher education institutions. Additional

policies issued by university administrators generally relate to the use of official LMS facilities.

In practice, widely used LMS platforms, such as Moodle, are full-featured. To support this policy, e-learning units have been established for various purposes, including e-learning management, training lecturers and students, and providing helpdesk services to resolve LMS-related issues.

According to an expert who serves as an LMS trainer, one of the obstacles in training users on the official LMS is the abundance of LMS features and the lengthy steps required to access them. This complexity makes it difficult to explain to users and raises doubts about the potential increase in the number of official LMS users. From a cognitive load perspective, the procedural tasks involved create cognitive pressure. This is supported by the expert's statement.

"This is another (different) cognitive process. That's why we need e-learning heroes to help selective attention of LMS training participants." (Interviewee MW, F).

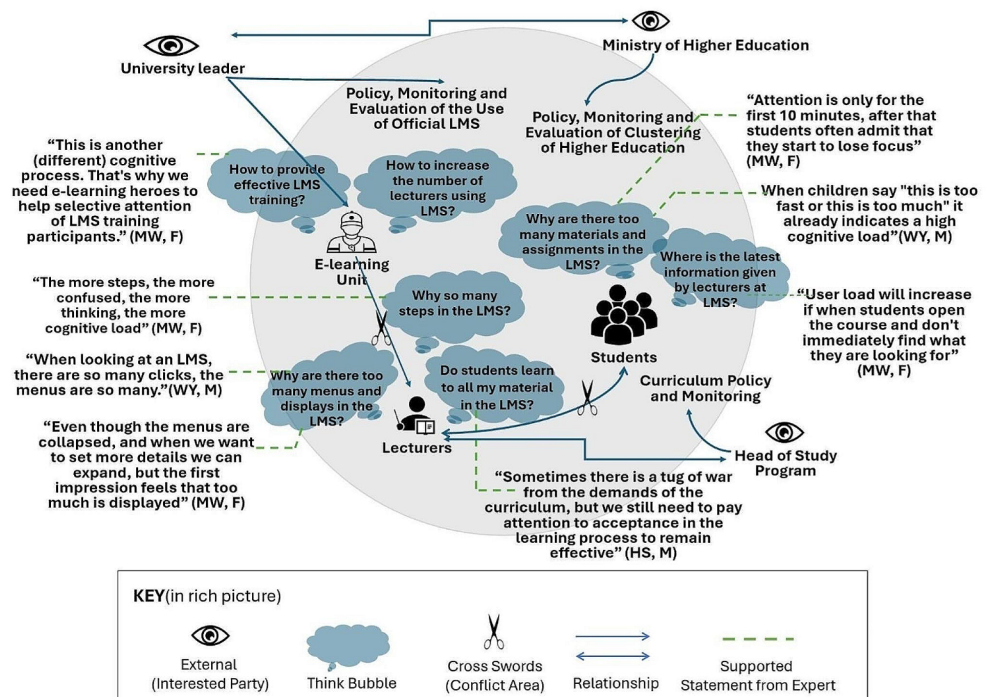
Other problems related to LMS were also conveyed by experts based on the lecturer's point of view, where the official LMS has long steps to achieve a goal and each step has many menu choices which make it confusing. This is supported by the following statements.

"The more steps, the more confused, the more thinking, the more cognitive load. Even though the menus are collapsed, and when we want to set more details we can expand, but the first impression feels that too much is displayed." (Interviewee.

MW, F)

, and.

Fig. 4 Rich picture related to the problem of using the official LMS by related stakeholders.
Source: Create by authors



“When looking at an LMS, there are so many clicks, the menus are so many.” (Interviewee WY, M).

Based on the results of interview triangulation in stage 1, the author develops a rich picture that provides a broad overview of the existing problems. An overall picture of user problems when using an LMS from the various stakeholders involved can be seen in the rich picture shown in Fig. 4. In the use of LMS, especially in Indonesia, there are policies, one of which is the Ministry of Education regarding the use of LMS as learning support. Later, the use of LMS will be evaluated and become one of the supporting points for higher education clustering. Then other derivative policies, issued by university administrators, generally relate to the use of official LMS facilities. In its implementation, the LMS that is widely used is a full-featured LMS such as Moodle. With this policy, an e-learning unit was formed for several purposes, such as e-learning management, training lecturers and students, to acting as a helpdesk to help solve problems related to LMS.

In addition to problems related to the use of LMS by lecturers, questions often arise from lecturers regarding whether students listen to all the material in the LMS and understand? The lack of monitoring and monitoring sessions in online learning in LMS often raises these doubts. Meanwhile, learning outcomes must still refer to curriculum policies. This is supported by the following expert statement:

“Sometimes there is a tug of war from the demands of the curriculum, but we still need to pay attention to acceptance in the learning process to remain effective” (Interviewee HS, M).

From the role of students, there are often complaints about the difficulty of finding the latest information from lecturers when there is a lot of material. This is supported by the statement:

“User load will increase if when students open the course and don’t immediately find what they are looking for” (Interviewee MW, F).

In addition, complaints from students regarding the amount of material presented in the LMS also indicate an increase in cognitive load. This is supported by expert statements as follows.

“Attention is only for the first 10 minutes, after that students often admit that they start to lose focus” (interviewee MW, F).

and.

“When children say, “this is too fast, or this is too much” it already indicates a high cognitive load” (Interviewee WY, M).

There are two conflict areas in the rich picture, but the main issue to be resolved is between lecturers and students. Lecturers feel the need to provide all materials that support learning, which can overwhelm students. Conversely, students try to fulfill their obligations to participate in the learning process but experience a high cognitive load, leading to overload or distraction. This problem needs to be addressed so that the LMS system can control and monitor the learning process based on the cognitive load capacity of the students while still achieving learning goals.

If the conflict between lecturers and students can be resolved through a system solution, it could also help address the issues faced by the e-learning unit. To narrow down the choice of solutions, the results of the rich picture analysis are considered in determining the root definition system in stage 3.

4.3 The result of stage 3: Root definition of relevant system

In determining the root definition of the relevant system, a benchmarking process was conducted by reviewing related literature. Three main sources served as references for system benchmarks, particularly starting from the user modeling process.

The first reference is the idea of an adaptive LMS system proposed by Kolekar et al. (2019), where the LMS system can adapt the UI using a rulebased method. In this case, the user profile or model is built from the user activity log, which is mapped into the Felder-Silverman Learning Style Model (FSLSM). Each category of the FSLSM is associated with an adaptive user interface that includes learning paths, learning objects, and additional learning resources. The adaptive process of this system is based on the implementation of predetermined rules.

The second benchmark system is proposed by Maaliw (2020). This LMS system adapts the UI and content based on mining learner log data to obtain learner behavior as a user model. One of the advantages of this system is the facility for visualization reports for lecturers or observers.

The third benchmark is from Lwande et al. (2021), which focuses on the cognitive detection aspect of using LMS. In addition to using activity log data for learning style mapping, this system also attempts to capture the user’s cognitive traits model.

From these three benchmarks, it was found that the primary modality in developing an adaptive LMS in terms of content and appearance is through activity logs, which are

mapped into the FSLSM and the Cognitive Traits Model. Students receive content and display recommendations, while lecturers can observe the student learning process through the visualization reports feature.

However, none of these benchmarks have incorporated the concept of cognitive load theory as a determinant of cognitive load. This includes categorizing working memory into intrinsic cognitive load, extraneous cognitive load, germane cognitive load, and sparse capacity.

The results of the benchmark process from the relevant system become the basis for answering the root definition keys, namely what, how, and why. The root definition obtained is:

“A system that can provide adaptive interfaces and learning strategies in an LMS by capturing the user’s cognitive load and combining with learner log data in order to jointly monitor learning according to each user’s abilities”.

Furthermore, the CATWOE analysis is carried out to support the root definition and provide an initial overview of the formulation of the proposed solution. The results of the CATWOE analysis are as follows:

- C (Client): Student.
- A (Actor): Lecturer.
- T (Transform): Transforming user model cognitive load based (input) to personal UI and learning strategies recommendation (output).
- W (World view): Understanding the user’s cognitive load (user model) when using the LMS system can support the system to be able to provide UI and learning strategy recommendations according to the user so that learning monitoring can be carried out.
- O (Owner): University Manager.
- E (Environment): Ministry of Education’s policy on the use of LMS in online/hybrid learning.

The results of the root definition and CATWOE analysis serve as the foundation and guide for developing the conceptual model. The conceptual model created in stage 4 represents a more concrete step toward the solution, which, in this case, is a system that can model LMS users based on cognitive load.

4.4 The result of stage 4: Conceptual model

The results of the root definition in stage 3 serve as the foundation for developing the conceptual model. During the preparation of the conceptual model, the interview results were thoroughly examined and triangulated to address each question related to capturing cognitive load from the

beginning to the end of learning in the LMS. This process allowed for drawing comprehensive conclusions.

The first interview question: “*What are the individual aspects that need to be considered in detecting a user’s cognitive load?*” This question received responses from three experts as follows.

- “Before detecting CL at the beginning of learning, it is necessary to check a person’s actions and reactions to the LMS, it can be called cognitive load or think about extraneous variables such as unfamiliar (prior knowledge). seeing the detection of his working memory then we know whether he is more easily distracted or not)” (Interviewee WY, M).
- “It is necessary to check motivation in using LMS, lecturers want it, let alone supported by university policies” (Interviewee MW, F).
- “... in detecting CL, not only individual differences (motivation, previous knowledge) but also in terms of the design of the task as well. We must be able to see where it contributed, whether due to changes in the design, different types of tasks or from individual differences” (Interviewee HS, M).

Based on the statements of the experts, three out of four experts stated that the individual aspects that need to be considered in detecting the user’s cognitive load are motivation, prior knowledge related to the LMS and materials, and working memory capacity.

The second interview question: “*How to measure a person’s cognitive load at the beginning before studying? What task is best to use?*” This question received responses from three experts as follows.

- “When it comes to technique, we usually look at Cognitive Load from observing students. For example, the most visible thing when we first ask a question is that someone responds. When asked in the middle of learning, the more information is given, the fuller the cognitive load” (Interviewee MW, F).
- “...check the load capability first without using the LMS and then use the LMS. Is the working memory before using the LMS high or low, if it is high, then using the LMS will be easy, but if it is low then it will likely have difficulty processing the LMS” (Interviewee WY, M). He also added a statement regarding the types of tasks that can be used to measure working memory capacity. “If you want to measure it, there are usually measurement tasks using span, there are digit spans forward and backward (audio/visual). Backward is talking about working memory, how flexible is it? the individual

performs the maintenance after that reproduces the requested, ...” (Interviewee WY, M).

- “Experimental studies usually have tasks that are carried out by respondents. However, it is necessary to specify what abilities you want to test. Then the interval/range/level of difficulty from one point to another point. Usually, here there are several domains such as attention, memory, it could also be towards visual search” (Interviewee RR, F).

Based on the experts’ statements, two of the four experts stated that at the beginning of learning, it is necessary to measure working memory capacity using visual or auditory tasks such as digit span forward/backward and visual search, or by observing students through questioning that is not directly related to the LMS context.

The third interview question: “*What are the techniques/methods for capturing cognitive load during the learning process?*” This question received responses from four experts as follows.

- “For measurement, it must be combined, if for self-report it is subjective, if you can use it directly to the brain it will be more visible, for example, it can be seen from the prefrontal cortex, if there is a distraction it will be visible, and if it can be compared with self-report and it’s possible if it’s on a lab scale” (Interviewee MW, F).
- “In the learning process, setting in an experiment, my understanding is that he gets multiple tasks at one time... many are memorized and cannot be ignored, then they are asked to remember the theories presented (performance). Yes, it’s his ability, not his perception, and it is more objective than a questionnaire” (Interviewee WY, M).
- “The research is made in two layers, for example, the first focuses on behavior first (accuracy and general psychology are more about time, intensity level, confidence/low confidence). We will investigate the second experimental layer related to EEG or ECG, eye tracking, or pupil dilation” (Interviewee RR, F).
- “Subjective are mixed with an objective. We will choose this objective again based on performance, it could be the length of time we work on (response time), for LMS, for example, from the movement of the mouse, how many times the mouse clicks, and so on (behavior)” (Interviewee HS, M).

Based on the statements of the experts, it can be concluded that three out of four experts believe that the method of capturing cognitive load in the learning process should use a combination of subjective and objective methods

(performance, behavior, and physiology) with research/experiments conducted in layers.

The fourth interview question: “*Is there a specific time reference for measuring cognitive load in the learning phases?*” This question received responses from four experts as follows.

- “There is a range of attention, ... if in psychology there is academic support, ideally it can last 20 minutes. ... An ideal basis for students could be to arrange 1 round of Pomodoro, do 1 activity window, then take a short break, then do 1 more activity and then long break” (Interviewee MW, F).
- “In flipped learning, based on research, he said it should not take more than 12 min. Video is the same as a presentation, right, it means that when we give the material, we can’t pass the time limit because later the child is too tired, so they don’t focus anymore on receiving the material” (Interviewee WY, M).
- “Usually, it is reproducibility from the previous research. For example, in terms of flow or the target being measured is the same, it’s best if we follow the flow, how many breaks are there in one performance” (Interviewee RR, F).
- “Usually, we already have standardization at the center if the video can’t be longer than 7 min” (Interviewee HS, M).

Based on the statements of the experts, there is no specific agreement regarding the appropriate period or duration for capturing cognitive load in the learning process. The suggested range varies between 7 and 20 min or refers to other studies that have been conducted.

The fifth interview question: “*In the context of using LMS, at which stages/cases are likely to increase cognitive load?*” This question received responses from two experts as lecturers who are more experienced in using LMS as follows.

- “In LMS (Moodle), activities with high CL, such as quizzes and lessons (the highest top) are followed by discussion forums, because it is more difficult to engage them. Engage in discussions can increase cognitive load compared to sending assignments” (Interviewee MW, F).
- “Example: the task of understanding video material. Video material is a complex task, from the start he must know to be able to see, identify the location of the material, then he opens, clicks, then he listens, listens and so on, it’s a relatively cognitive process complex” (Interviewee HS, M).

Based on the statements of the experts, two out of four experts stated that the activities in the LMS (Moodle-based) are in the order of the highest cognitive load (1) Lesson (material) and Quiz, (2) Discussion forums (3) Assignment. However, this is still broadly speaking, each navigation or procedural task to achieve the objectives of these activities needs to be further detailed.

The sixth interview question: *“From the UI side, do color, layout, and text properties affect cognitive load?”* This question is an initial question to gain insight into the implications of developing user interfaces later. This question received responses from all experts as follows.

- “...there is an impact on color on learning” (Interviewee MW, F).
- “Color psychology doesn’t affect it, because color depends on one’s interest, it might affect people who like or don’t like certain colors. Usually, the colors used are pastel tones, they are more general. Text properties are more simplistic, and make it easier to read, giving focus attention ” (Interviewee WY, M).
- “Color is subjective, maybe there are people who have a favorite color, more responsive if they see a certain color that they like. The color direction userfriendly or not. The clearer the display, the easier self-navigation for the interface, it’s not too small and it’s too big. This could be a point that we research for the ideal font size, it could be research not just an opinion” (Interviewee RR, F).
- “But in my opinion, it will be very influential (text properties). For example, for the case of reading comprehension, research on the use of romance and non-romance will have an effect” (Interviewee HS, M).

Based on the statements of the experts, three out of four experts stated that color did not affect cognitive load, while text properties did. The results of interview triangulation from the six questions related to the process of capturing cognitive load in the LMS are mapped into the conceptual model in Fig. 5 as follows.

Based on the conceptual model regarding the process of capturing cognitive load during the learning process, it is obtained that the extraction of possible tasks and data can be used as user profiles. These data can be used as a possible dataset that models the user for the adaptive engine process before giving design implications and appropriate material recommendations. Next, Fig. 6 provides an overview of the initial cognitive load-based LMS user model design.

Based on the conceptual model for capturing cognitive load during the learning process, it was determined that the extraction of possible tasks and data can be used to create user profiles. These data can form a dataset to model the user for the adaptive engine process, which will then inform design implications and appropriate material recommendations. Figure 6 provides an overview of the initial cognitive load-based LMS user model design.

Motivation and prior knowledge need to be assessed before measuring the user’s working memory capacity. In the proposed model, learning motivation is

typically measured using a questionnaire, such as the Motivated Strategies for Learning Questionnaire (MSLQ) developed by Pintrich and De Groot (1990). Similar questionnaires can also be used to assess motivation. Additionally, prior knowledge can be evaluated through assessments tailored to the specific material or course.

Fig. 5 Conceptual model for capturing user’s cognitive load while using the learning management system. *Source:* created by authors

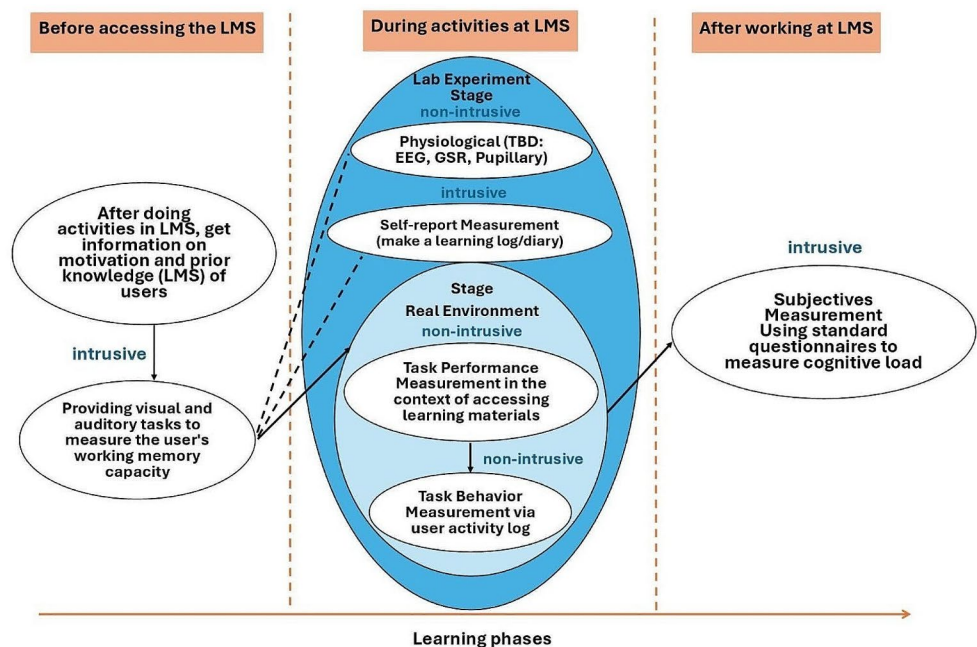
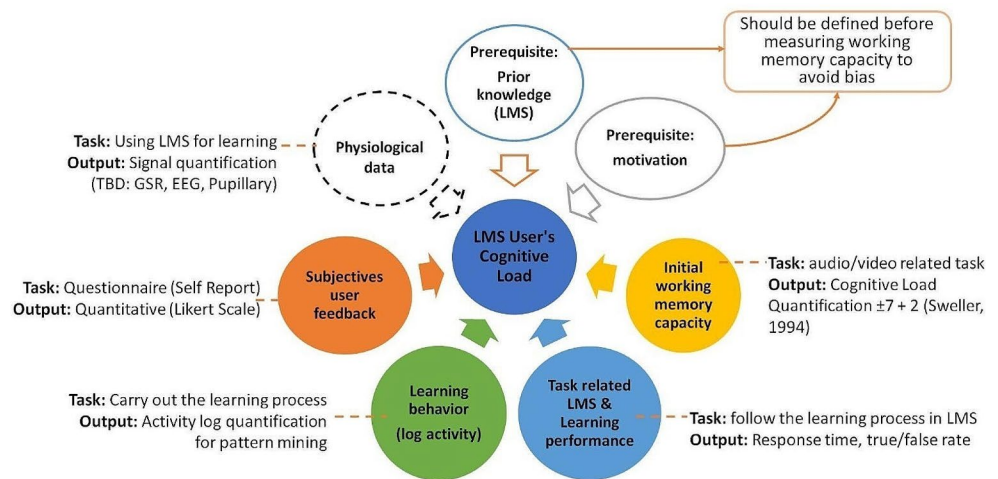


Fig. 6 An initial design of the LMS user model based on cognitive load. *Source:* created by authors



According to experts, in a cognitive load-based LMS, motivation and prior knowledge play crucial roles in optimizing user cognitive load. These elements can be interconnected and combined to enhance cognitive load management. High motivation can help reduce cognitive load by increasing users' focus and engagement in learning. Motivated users tend to absorb and process information better, thus reducing the cognitive pressure associated with learning. Utilizing prior knowledge aims to acknowledge and leverage users' existing knowledge to optimize learning. By building on users' prior knowledge, learning materials can be presented more relevantly and understandably, reducing the cognitive load of assimilating new information. Information about motivation and prior knowledge can provide additional insights into fluctuations in cognitive load when users access the LMS.

The user's working memory capacity needs to be measured before using the LMS. Initially, data obtained to model the user include prior knowledge, motivation, and working memory capacity. After entering the LMS system, performance data (response time, true/false rate), behavior data (activity log), and self-report results (Likert scale) can be captured. Physiological data can be obtained in a laboratory setting.

At this stage, the research question has been answered. The conceptual model developed to capture the user load when using the LMS is then compared with the problem situation to assess the possible challenges when implementing the conceptual model. The details of the comparison can be seen in the results of stage 5.

4.5 The result of stage 5: Comparing conceptual model with the problem situation - model validation

At stage 5, the conceptual model from stage 4 is criticized by comparing every aspect of its implementation with the problem situation identified in stage 2. This stage is also

referred to as the model validation stage. The validation stage involves the participation of four experts who provide feedback on the previous stages. The model validation process is conducted using a formal questioning technique, which involves asking in-depth questions about the model to the experts. By using the formal questioning method, this stage focuses on discussing points that might become conflicts if the user modeling process and cognitive load detection are implemented in the real world.

4.5.1 First iteration

There were three follow-up questions posed to the experts with the following responses.

The first follow-up question: *"How to provide a visual or auditory task to measure working memory capacity before using the LMS?"* The expert responses are:

- "In the experimental setting, we consider the simplicity of what we want to measure, and in the experimental setting, maybe 1 display is not rich in story or plot, but rather towards repetition conditions or appearance in this case" (Interviewee RR, F).
- "In cognitive terms, it usually depends on the theme, the meaning depends on the aims of the research. But the simplest one is giving Stroop tasks, letter and number sequences, digit span forward-backward in visual and auditory form" (Interviewee WY, M).

The second follow-up question: *"How is the capturing cognitive load in the learning process?"* The responses obtained are.

"In order to capture cognitive load in learning process, participants can be asked to take notes at what time of class they are attending. Then, at what minute

they are absent, asked to write a diary entry during the lecture on Zoom. Meanwhile, in LMS, a diary can be used if the learning setting is fully online, for example, MOOC, and it is quite hard to use for the hybrid. The diary can be biased, or it has to be on cam. LMS must be able to capture what they are studying. In research, participants can be observed, so they can use the lab settings first” (Interviewee MW, F).

Finally, the third follow-up question: “*How can capturing cognitive load.*

(CL) be in line with the learning material?” The response from the expert is.

“We don’t design specifically what parts are given so that the CL becomes smaller than the CL can be large. But usually, we are aware that checking feedback is done. ...but maybe other approaches need to be considered when preparing the lesson plan. Practically, we do not select what kind of CL should be used, but directly integrate the CL as part of a comprehensive learning process including aspects of individual characteristics, motivation, and also those related to organizing knowledge in participants” (Interviewee HS, M).

The process of comparing conceptual models and problem situations focuses on how cognitive load can be captured at each point in the learning process. This is because, based on information from experts, teachers often do not allocate specific time to check the cognitive load of their students during the learning process. Therefore, when a system is specifically implemented to check cognitive load, a strategy is needed to guide the capturing process.

Based on insights from experts, there are three points that need to be considered. First, to capture cognitive load before using the LMS, simple and specific tasks commonly used in psychology can be employed. This ensures that working memory is not influenced by prior familiarization with the LMS. Second, during the learning process, a scenario for learning activities is needed to capture cognitive load. Due to the complexity of the learning process, additional variables will be introduced when measuring cognitive load. A specific learning scenario should be designed, and then the learning activities should be mapped to the user’s cognitive load state. Third, the correlation between capturing cognitive load and the learning materials should be considered. The materials need to be designed based on cognitive load theory, particularly the concept of element interactivity.

4.5.2 Second iteration

In the second iteration, still with the same experts, we delve deeper into the stages of the model, tasks, and the generated data. The validated model results (Fig. 6), agreed upon by the experts, are as follows:

1. In the initial stage before using the LMS, there may be other factors besides motivation and prior knowledge, such as learning preferences, that need to be considered. These additional factors need to be added as adjustable components to assess the initial conditions of users.
2. During the learning phase, it is necessary to clarify the contexts of learning phases, tasks, generated data, and the possibility of the model providing options for unimodal and multimodal approaches in detecting cognitive load, as presented in Fig. 5.
3. Clarifying the model by adding a legend and other additional information.

The results of this comparison are then further explored in stage 6, focusing on the possibility of implementing feasible solutions to detect cognitive load based on the user’s initial model.

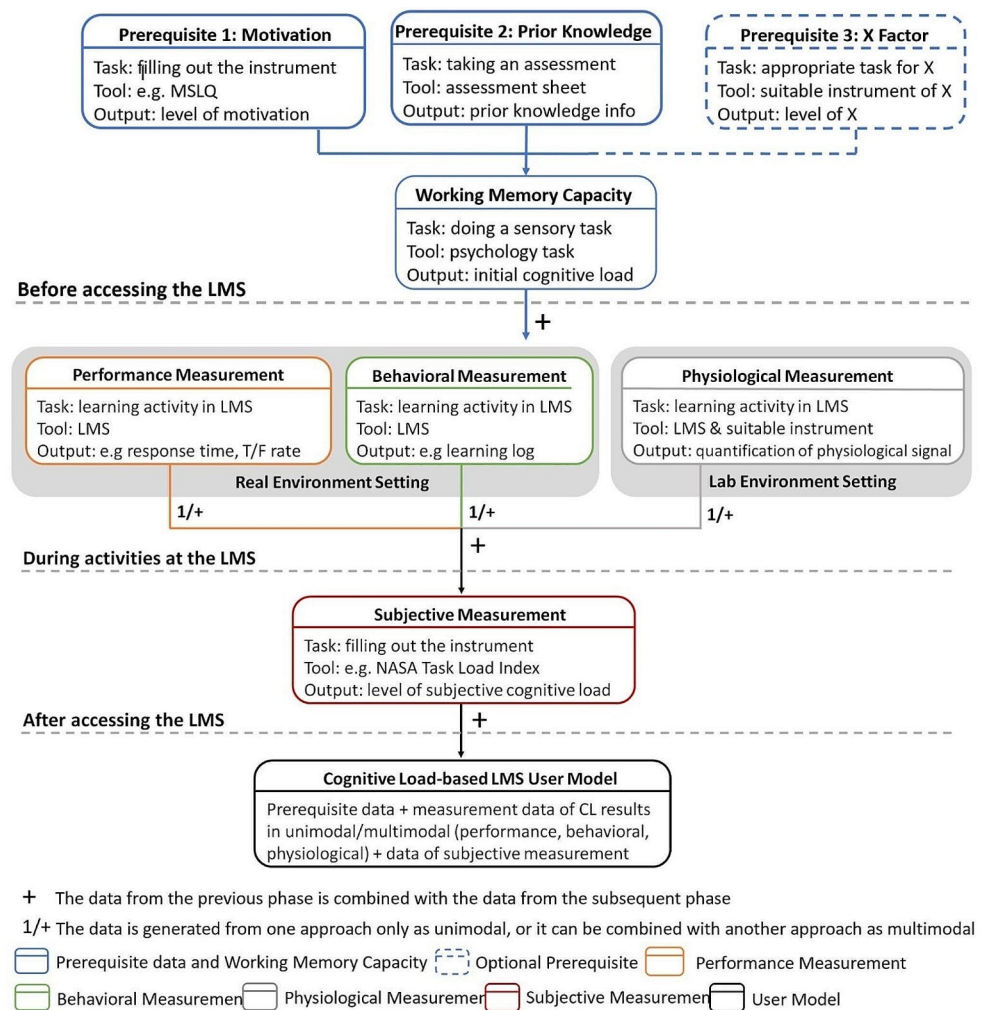
4.6 The result of stage 6: Feasible and desirable changes

At this stage, the discussion about feasible and desirable changes is a continuation of the results from the formal questioning in the previous comparison stage. After two iterations of model validation, the model improvements agreed upon by the experts can be seen in Fig. 7.

The prerequisite data needed before accessing the LMS can be adjusted according to the research context’s objectives. However, the minimum required data includes motivation and prior knowledge. It is possible to give a visual task to measure working memory capacity at the beginning of the learning process. This aligns with the research by Meyerhoff and Papenmeier (2020) regarding visual object tracking using PsychoPy tools. The flexibility to design psychological tasks can be combined with related user interface tasks (such as accessing menus, viewing information, using buttons, etc.).

As the learning process progresses, cognitive load is influenced not only by the user interface but also by the learning content. It is necessary to diversify methods for measuring cognitive load, one of which is learning analytics (Lang et al. 2022). Learning analytics involves the measurement, collection, analysis, and reporting of data about learners and their context, with the aim of understanding and optimizing learning and the environment in which it occurs.

Fig. 7 The revised of initial cognitive load-based LMS user model. *Source:* Created by authors



The data collected can be multimodal, including logs of learner interactions and environmental factors.

However, to bridge the gap between the possible implementation of the solution and the problem situation, it is necessary to consider the steps that can be taken as implications for implementing the system solution. The actions to improve the problem situation, in order to capture cognitive load and get initial user model proofing, are described in stage 7.

4.7 The result of Stage 7: Action to improve the problem situation (implication) and discussion

The last step of the SSM is to determine the actions that can be taken to improve the problem situation. It is hoped that the proposed solution that has been processed up to stage 6 can be implemented with a gap that is not too far from the real condition. To determine action, triangulation of expert information and supporting theory creates solution action. The details of stage 7 are as follows. Based on the expert's statement:

“What can be detected at the beginning is usually how good his working memory is. So, when we look at his working memory detection, we know whether he is more easily distracted or not, which is a cognitive load problem” (Interviewee WY, M).

This is supported by a study of latent variables examining whether different classes of working memory tasks measure the same general construct of working memory capacity (WMC) (Wilhelm et al. 2013). The implication is that it is necessary to arrange detailed tasks to measure the user's working memory capacity and can be implemented in the context of using LMS.

Another focus is related to efforts to capture the user's cognitive load during the learning process, both related to the user interface and its relevance to the material being studied. Based on the expert's statement.

“LMS must be able to capture learning while doing what. In research, participants can be observed” (Interviewee MW, F).

This is possible if we reflect on studies in the education domain. Data related to learning can be obtained from learner log data (TELE), sensory data, surveys, and interviews (Lang et al. 2022). This multimodal approach can strengthen the context of user behavior and learning. To capture cognitive load, researchers need to combine data derived from subjective (self-report) and objective (behavior, performance, physiological) sources to obtain a more comprehensive representation of the user model for precise mapping into intrinsic, extraneous, and germane cognitive load.

Based on our study of relevant literature over the past five years, there has not been a cognitive load-based LMS user model that detects or informs about the status or fluctuations of cognitive load when users engage in learning activities on an LMS. Successfully detected cognitive load status then becomes a trigger to determine LMS interface adaptations that can optimize user cognitive load. A similar study that approached this research is the study by Ho et al. (2019). They utilized the Multiple Criteria Decision Making (MCDM) analysis method of Grey Relational Analysis (GRA) and Reachable Grey Structural Modeling (RGSM) to identify the effective factors of users' cognition dimensions. The model produced a ranking level table and relative structural chart for users' cognition attitudes towards the LMS. However, Ho's research is different from the cognitive load-based LMS user model proposed in this study, where fluctuations in cognitive load can be measured multimodally, combining both subjective and objective measurements.

From the research that has been done, there are several limitations, including the fact that the SSM Model 1 method used has only been implemented in one iteration, so there are still many aspects that have not been addressed. In addition, the problems of new learners arise from the point of view of experts (lecturers) rather than from the perspective of direct students. Furthermore, the context for capturing cognitive load does not yet have a specific reference. Finally, there has been no in-depth study of the individual differences involved in the process of using e-learning, although experts emphasize the priority of motivation and prior knowledge.

5 Conclusion and future works

Several components need to be considered to detect cognitive load: prior knowledge of using LMS, motivation, and working memory capacity. Tasks and methods to measure the user's cognitive load should employ a mixed-method approach, combining subjective and objective measures (performance, behavioral, and physiological) captured during the learning process. A user model based on cognitive load can be created by combining working memory

capacity, task performance (time, true/false rate), learning behavior (learner log), physiological data (in a lab setting), and self-reporting.

This research is a preliminary study in designing a cognitive load-based LMS user model. In the future, research should focus on the following areas: Deepening the study iteratively with SSM Model 2 to obtain more detailed aspects. Enhancing the preparation of tasks to capture working memory capacity before learning begins. Compiling detailed performance measurements. Extracting learner data logs for measuring behavior. Determining which questionnaires will be used for self-report (subjective) measurement at the end of learning. Additionally, adaptive engine exploration should be conducted to define a machine learning architecture that fits the research context. Factorial experiments should be carried out to develop LMS interface designs that align with the user's cognitive load.

Author contributions Suryani conducted interviews with cognitive experts and wrote the main manuscript text. Santoso composed scenario interviews. Suryono checked the level of similarity of manuscript text. All authors reviewed the manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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